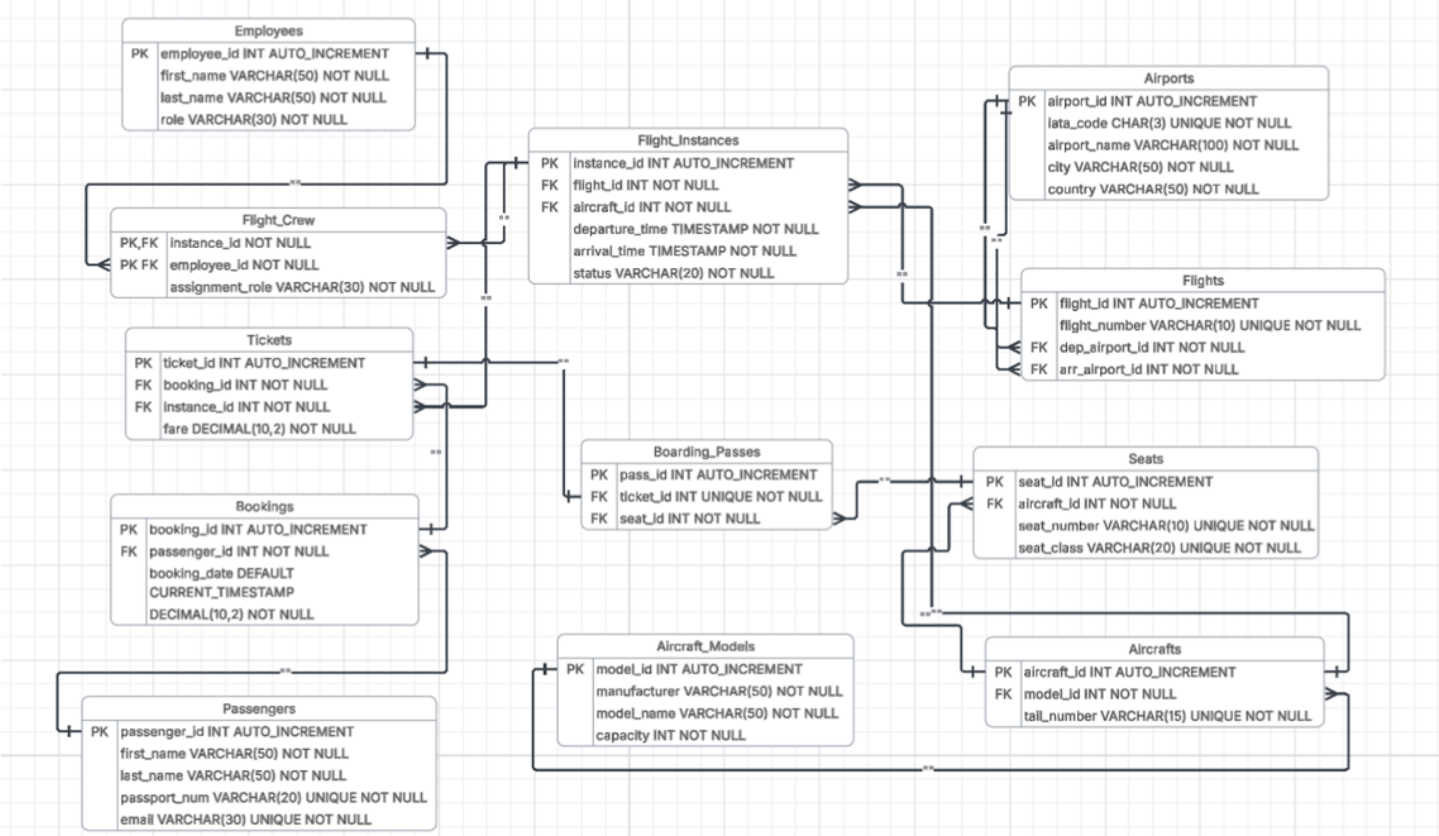
1.1 Logical Scheme



1.2 Objects

Table 1 — Airports

A comprehensive registry of all airports within the system, storing essential identification codes and location information for each facility.

Main Connections:

One-to-Many with flights (as departure airport).One airport can be departure point for multiple flight routes.

One-to-Many with flights (as arrival airport).One airport can be the arrival point for multiple flight routes.

Field Name	Field Description	Data Type / Constraints
airport_id	Unique identifier of the airport	INT, PK, AUTO_INCREMENT
iata_code	3-letter IATA airport code (e.g., "JFK")	CHAR(3), UNIQUE, NOT NULL
airport_name	Full name of the airport	VARCHAR(100), NOT NULL
city	City where the airport is located	VARCHAR(50), NOT NULL
country	Country where the airport is located	VARCHAR(50), NOT NULL

airport_id	iata_code	airport_name	city	country
1	JFK	John F. Kennedy International Airport	New York	USA
2	LHR	London Heathrow Airport	London	United Kingdom

Table 2—Flights

A catalog of all flight routes operated by the airline, defining the connection between departure and arrival airports without specific timing.

Main Connections

Many-to-One with Airports (departure) — each flight route has one designated departure airport.

Many-to-One with Airports (arrival) — each flight route has one designated arrival airport.

One-to-Many with Flight_Instances — one flight route can have multiple scheduled instances over time.

Field Name	Field Description	Data Type / Constraints
flight_id	Unique identifier of the flight route	INT, PK, AUTO_INCREMENT
flight_number	Public flight number (e.g., "KS123")	VARCHAR(10), UNIQUE, NOT NULL
dep_airport_id	Departure airport reference	INT, FK (Airports.airport_id), NOT NULL
arr_airport_id	Arrival airport reference	INT, FK (Airports.airport_id), NOT NULL

flight_id	flight_number	dep_airport_id	arr_airport_id
101	KS123	1	2
102	BA456	2	1

Table 3—Flight_Instances

The operational heart of the system that schedules a specific flight route with a specific aircraft at a set date and time, tracking real-time status.

Main Connections

Many-to-One with Flights — each instance is linked to one flight route.

Many-to-One with Aircrafts — each instance uses one specific aircraft.

One-to-Many with Tickets — one flight instance can have multiple tickets sold for it.

One-to-Many with Flight_Crew — one flight instance can have multiple crew members assigned to it.

Field Name	Field Description	Data Type / Constraints
instance_id	Unique identifier for a specific flight occurrence	INT, PK, AUTO_INCREMENT
flight_id	Flight route reference	INT, FK (Flights.flight_id), NOT NULL
aircraft_id	Aircraft used for this flight	INT, FK (Aircrafts.aircraft_id), NOT NULL
departure_time	Scheduled departure date and time	TIMESTAMP, NOT NULL
arrival_time	Scheduled arrival date and time	TIMESTAMP, NOT NULL
status	Current flight status (e.g., On Time, Delayed)	VARCHAR(20), NOT NULL

instance_id	flight_id	aircraft_id	departure_time	arrival_time	status
5001	101	201	2024-06-15 08:00:00	2024-06-15 16:30:00	On Time
5002	102	202	2024-06-15 14:00:00	2024-06-15 22:45:00	Delayed

Table 4—Aircraft_Model

A lookup table (dictionary) that stores aircraft model specifications, including manufacturer details and passenger capacity.

Main Connections:

One-to-Many with Aircrafts — one aircraft model can be used by many different physical aircraft

Field Name	Field Description	Data Type / Constraints
model_id	Unique identifier of the aircraft model	INT, PK, AUTO_INCREMENT
manufacturer	Manufacturer of the model (e.g., "Boeing")	VARCHAR(50), NOT NULL
model_name	Model name (e.g., "737-800")	VARCHAR(50), NOT NULL
capacity	Total passenger capacity	INT, NOT NULL

model_id	manufacturer	model_name	capacity
1	Boeing	737-800	189
2	Airbus	A320	180

Table 5—Aircrafts

An inventory of all physical aircraft in the fleet, linking each plane to its model specifications through registration numbers.

Main Connections:

Many-to-One with Aircraft_Models — each aircraft follows one model specification.

One-to-Many with Flight_Instances — one aircraft can be scheduled for multiple flights.

One-to-Many with Seats — one aircraft contains multiple seats.

Field Name	Field Description	Data Type / Constraints
aircraft_id	Unique identifier of the physical aircraft	INT, PK, AUTO_INCREMENT
model_id	Aircraft model reference	INT, FK (Aircraft_Models.model_id), NOT NULL
tail_number	Registration number of the aircraft	VARCHAR(15), UNIQUE, NOT NULL

aircraft_id	model_id	tail_number
201	1	N12345
202	2	G-ABCD

6—Employees

A personnel directory containing names and professional roles of all staff members employed by the airline.

Main Connections:

One-to-Many with Flight_Crew — one employee can be assigned to multiple flight instances.

Field Name	Field Description	Data Type / Constraints
employee_id	Unique identifier of the employee	INT, PK, AUTO_INCREMENT
first_name	Employee first name	VARCHAR(50), NOT NULL
last_name	Employee last name	VARCHAR(50), NOT NULL
role	Job role (e.g., Pilot, Flight Attendant)	VARCHAR(30), NOT NULL

employee_id	first_name	last_name	role
10	Sarah	Mitchell	Pilot
11	James	Anderson	Flight Attendant

Table 7—Flight_Crew

A junction table that manages the Many-to-Many relationship between employees and flight instances, acting as the official crew assignment list for every flight.

Main Connections:

Many-to-Many between Employees and Flight_Instances.

Uses a Composite Primary Key (instance_id, employee_id).

Both fields act as Foreign Keys.

Field Name	Field Description	Data Type / Constraints
instance_id	Flight instance reference	INT, FK (Flight_Instances.instance_id), PK (instance_id, employee_id)
employee_id	Employee reference	INT, FK (Employees.employee_id), PK (instance_id, employee_id)
assignment_role	Specific role for this flight (e.g., Captain)	VARCHAR(30), NOT NULL

instance_id	employee_id	assignment_role
5001	10	Captain
5001	11	Senior Flight Attendant

Table 8_Passengers

The central repository for all registered passengers, storing their personal details and travel documentation for booking purposes.

Main Connections:

One-to-Many with Bookings — one passenger can make multiple bookings.

Field Name	Field Description	Data Type / Constraints
passenger_id	Unique identifier of the passenger	INT, PK, AUTO_INCREMENT
first_name	Passenger first name	VARCHAR(50), NOT NULL
last_name	Passenger last name	VARCHAR(50), NOT NULL
passport_num	Passport number	VARCHAR(20), UNIQUE, NOT NULL
email	Passenger email address	VARCHAR(100), UNIQUE

passenger_id	first_name	last_name	passport_num	email
301	Emily	Johnson	P12345678	emily.j@email.com
302	Michael	Brown	P87654321	m.brown@email.com

Table 9—Bookings

A financial and reservation log that records every booking transaction made by passengers, tracking total amounts and booking timestamps.

Main Connections:

Many-to-One with Passengers — each booking belongs to one passenger.

One-to-Many with Tickets — one booking can contain multiple tickets.

Field Name	Field Description	Data Type / Constraints
booking_id	Unique identifier of the booking	INT, PK, AUTO_INCREMENT
passenger_id	Passenger who made the booking	INT, FK (Passengers.passenger_id), NOT NULL
booking_date	Date and time of the booking	TIMESTAMP, DEFAULT CURRENT_TIMESTAMP
total_amount	Total price for the booking	DECIMAL(10,2), NOT NULL

booking_id	passenger_id	booking_date	total_amount
4001	301	2024-05-20 10:30:00	850.00
4002	302	2024-05-21 14:15:00	1200.50

Table 10—Tickets

Individual flight tickets that link bookings to specific flight instances, storing fare information for each segment of travel.

Main Connections:

Many-to-One with Bookings — each ticket belongs to one booking.

Many-to-One with Flight_Instances — each ticket is linked to one flight instance.

One-to-One with Boarding_Passes — each ticket generates one boarding pass.

Field Name	Field Description	Data Type / Constraints
ticket_id	Unique identifier of the ticket	INT, PK, AUTO_INCREMENT
booking_id	Booking reference	INT, FK (Bookings.booking_id), NOT NULL
instance_id	Flight instance reference	INT, FK (Flight_Instances.instance_id), NOT NULL
fare	Price for this specific ticket	DECIMAL(10,2), NOT NULL

ticket_id	booking_id	instance_id	fare
7001	4001	5001	850.00
7002	4002	5002	1200.50

Table11—Seats

An inventory of all seat configurations within each aircraft, defining seat numbers and class categories for passenger assignment.

Main Connections:

Many-to-One with Aircrafts — each seat belongs to one aircraft.

One-to-Many with Boarding_Passes — seats can be assigned to passengers across flights.

Field Name	Field Description	Data Type / Constraints
seat_id	Unique identifier of the seat on an aircraft	INT, PK, AUTO_INCREMENT
aircraft_id	Aircraft reference	INT, FK (Aircrafts.aircraft_id), NOT NULL
seat_number	Seat designation (e.g., "12A")	VARCHAR(5), NOT NULL
seat_class	Class of the seat (e.g., Economy, Business)	VARCHAR(20), NOT NULL

seat_id	aircraft_id	seat_number	seat_class
8001	201	12A	Economy
8002	201	1A	Business

Table 12—Boarding_Passes

The final travel document that assigns a specific seat to a ticket holder, enabling passenger boarding for their flight.

Main Connections:

One-to-One with Tickets — each boarding pass corresponds to one ticket.

Many-to-One with Seats — each boarding pass assigns one seat.

Field Name	Field Description	Data Type / Constraints
pass_id	Unique identifier of the boarding pass	INT, PK, AUTO_INCREMENT
ticket_id	Ticket reference (one-to-one)	INT, FK (Tickets.ticket_id), UNIQUE, NOT NULL
seat_id	Assigned seat reference	INT, FK (Seats.seat_id), NOT NULL

pass_id	ticket_id	seat_id
9001	7001	8001
9002	7002	8002